Unit II: Plant Morphology and Taxonomy of Angiosperm

Chapter 3

Vegetative Morphology

Learning Objectives

The learner will be able to,

- *Explore the parts of the flowering plants*
- *Differentiate vegetative morphology and reproductive morphology*
- *Compare various Root systems and their modifications*
- *Understand the stem modifications and functions*
- *Interpret the structure and functions of leaf*

Chapter Outline

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3.1 Habit
3.2 Habitat
3.3 Life Span
3.4 Parts of a flowering plant
3.5 Root System
3.6 Shoot system
3.7 Leaf

The study of various external features of the organism is known as **Morphology. Plant Morphology** is also known as **external morphology that** deals with the study of shape, size and structure of plants and their parts (roots, stems, leaves, flowers, fruits and seeds). Study of Morphology is important in Taxonomy. Morphological features are important in determining productivity of crops. Morphological characters indicate the specific habitats of living as well as the fossil plants and help to correlate the distribution in space and time of fossil plants. Morphological features are also significant for phylogeny.

Plant Morphology can be studied under two broad categories:

A. **Vegetative Morphology** – It includes shoot system and root system
B. **Reproductive Morphology** – It includes Flower/Inflorescence, Fruit and Seed

A. Vegetative morphology

Vegetative morphology deals with the study of shape, size and structure of plants and their parts like roots, stems and leaves. To understand the vegetative morphology the following important components are to be studied. They are, 1) Habit, 2) Habitat and 3) Lifespan.

3.1 Habit

The general form of a plant is referred as habit. Based on habit, plants are classified into herbs, shrubs, climbers (vines) and trees.

I. Herbs

Herbs are soft stemmed plants with less wood or no wood. Example: *Phyllanthus amarus, Cleome viscosa.* According to the duration of their life they may be classified as **annuals**, **biennials** and **perennials**. Perennial herbs having a bulb, corm, rhizome or tuber as the underground stem are termed as **geophytes.** Example: *Allium cepa*

II. Shrubs

A shrub is a perennial, woody plant with several main stems arising from the ground level. Example: *Hibiscus rosa-sinensis* (shoe flower)

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